Studying Functional Brain Networks to Understand Mathematical Thinking: A Graph-Theoretical Approach

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Abstract—**Brain function during mathematical thinking is a common concern of scientists from different research fields. The study of functional brain networks extracted from electroencephalographic (EEG) signals using graph theory seems to meet the challenge of neuroscience to understand brain functioning in terms of dynamic flow of information among brain regions. Some studies have found differences between the basic arithmetic operations among brain regions; however, they have not explained the brain function during difficult mathematical tasks and complex mathematical processing. This study investigates the changes of the functional networks' organization among different mathematical tasks. To this end, EEG data from 10 subjects were recorded during three different tasks, Number Looking which was served as the control situation, Simple Addition which refers to the addition of single-digit numbers and Difficult Multiplication with trials of two-digit multiplications. We analyzed weighted graphs so as to provide a more realistic representation of functional brain networks. Mutual information was employed to form the weights among the different channels, while various global and local graph indices were further examined. The results suggest that there are some statistically significant differences between graph theoretical indices among the different tasks and their range of values depend on the particular task.**

Keywords—**Mathematical Cognition, Graph Theory, Mathematics, Mutual Information, Functional Connectivity.**

I. INTRODUCTION

Numerical estimation, commonly known as subitizing, refers to the rapid and accurate judgments of number without actual counting. Enumeration of elements results from various cognitive processes and involves different brain regions. The most important components of this system are the cortical structures located in the posterior parietal and parieto-occipital region [1-4].

Left angular gyrus (lAG) seems to incorporate a crucial component of the system which accounts for learning and retrieving math facts [5-7]. According to a recent study [8] the lAG is involved in the retrieval of verbally stored arithmetic facts (such as multiplication facts or simple additions) by operating within the language system through its connections to the left temporo-parietal cortex [9]. Another study found that that lAG is engaged in arithmetic fact retrieval together with frontal structures [10]. Moreover, this region appears to be more active in exact calculation. During exact calculation the lAG shows higher activation for arithmetical operations related to rote verbal memory arithmetic facts. Thus, lAG shows more activation for multiplication tables in relation to subtraction or number comparison [8, 11].

Some studies have found differences between the basic arithmetic operations among brain regions apart from those who focused predominately on the frontal and parietal lobes. However, they have not explained the brain function during difficult mathematical tasks and complex mathematical processing. In fact, what are the series of events in which the brain does advanced mathematical processing and what is the exact mechanism that supports mathematics like fractions, complex numbers and calculus are difficult questions to answer. Nonetheless, Intraparietal sulcus (IPS) is a key region for mathematics which is activated when numbers are presented in various notations or spoken number words. In addition to IPS, parietal frontal networks, some basal ganglia and other networks associated with long-term and working memory, are engaged in difficult problem solving.

A variety of problems that concern brain function can be dealt with network comparison and characterization. The problem of comparing these networks is the task of measuring their topological difference or similarity. In order to investigate the changes of the functional networks' organization among different tasks (number looking, simple additions and two digit multiplications), we analyzed weighted graphs in which the weights were extracted using a nonlinear association measure. To this end, functional networks were extracted from EEG signals, and then we computed global and local graph theoretical indices for the networks so as to investigate if their values depend on different tasks.

II. MATERIALS AND METHODS

A. Participants

Ten right handed medical students of the University of Crete were recruited for this study. Participants with a history of psychiatric or neurological illness or under

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medication were excluded from the study. All participants had normal or corrected to normal vision. Recordings were performed during the morning, while all participants signed an informed consent form, while the experimental protocol was approved by the ethics committee of the Department of Medicine of the University of Crete.

B. Experimental Design

EEG data were acquired during three different tasks, Number Looking (NL) which was served as the control situation and contained 20 trials, Simple Addition (SA) which refers to the addition of single-digit numbers (e.g. 3+8) with 20 trials as well, and Difficult Multiplication (DM) with 8 trials of two-digit multiplications (e.g. 32x75). All the responses for the calculation tasks were acquired orally. For the purpose of the current study, we chose pieces of 10s duration, without visible artifacts and with correct answers and all the signals were further cleared by ocular artifacts using the REGICA [12] plugin for the EEGLAB toolbox [13]. No response was required during number looking and fixation, while for the SA and DM task students were not restricted to a prescribed time period to provide their answer.

C. Functional Connectivity

Mutual Information index is given by the next formula:

$$I(X;Y)=\sum_{y\in Y}\sum_{x\in X}p(x,y)\log\frac{p(x,y)}{p(x)\,p(y)}$$

where $p(x,y)$ is the joint probability distribution and $p(x)$, $p(y)$ are the marginal probability distribution functions of X and Y respectively.

The functional connectivity was computed using a normalized variant of mutual information:

$$\frac{I(X;Y)}{\sqrt{I(X)\cdot I(Y)}}$$

where $I(X)$, $I(Y)$ are the entropies [14] of X and Y respectively.

D. Graph Analysis

A graph is a pair of two sets, $G=(V,E)$ where V is a set of objects called nodes and E is a set of a collection of pairs of these objects which are connected with links. In a weighted graph, each link is associated with a number called the weight. In our study, the nodes are the channels and the weights are defined by the value of mutual information of each pair. We constructed 30 (3 tasks x 10 subjects) weighted brain networks from the correlation matrices.

In this study, we used six networks indices which are described next.

1. Clustering Coefficient:

Clustering coefficient is a measure of degree to which vertices in a graph tend to cluster together. Onnela et al [15], introduce subgraph "intensity" as the geometric mean of its link weights and "coherence" as the ratio of the geometric to the corresponding arithmetic mean. Using these measures, motif scores and clustering coefficient can be generalized to weighted networks. Clustering coefficient is given by the following type:

$$c^{w}=\frac{1}{n}\sum_{i\in N}\frac{2t_i^W}{k_i\left(k_i-1\right)}$$

where $t_i^W=\frac{1}{2}\sum_{j,h\in N}\left(w_{ij}w_{ih}w_{jh}\right)^{\frac{1}{3}}$ is the (weighted) geometric mean of triangles around.

2. Global Efficiency

Efficiency of the path between two nodes is the inverse of their shortest distance. In case where there is not any path to connect two nodes, their distance is infinite so their efficiency is 0. Global efficiency (GE) [16,17,19] is given by the following formula:

$$E^W=\frac{1}{n}\sum_{i\in N}\frac{\sum_{j\in N,\,j\neq i}\left(d_{ij}^W\right)^{-1}}{n-1}$$

where N is the number of vertices in graph G and $d_{ij}^W=\sum_{a_{uv}\in g_{i\leftrightarrow j}^{w}}f\left(w_{uv}\right)$, where f is a map from weight to length and $g_{i\leftrightarrow j}^W$ is the shortest weighted path between i and j.

3. Modularity

The modularity [18] is a statistic that quantifies the degree to which the network may be subdivided into clearly delineated groups (modules) and is given by the next formula:

$$Q^W=\frac{1}{l^W}\sum_{i,j\in N}\left[w_{ij}-\frac{k_i^W k_j^W}{l^W}\right]\delta_{m_i,m_j}$$

where k_i^W is the strength of node i, l^W is the sum of all weights in the network, m_i is the module containing the node i and $\delta_{m_i,m_j} = 1$ if $m_i = m_j$ or 0 otherwise.

4. Strength

Strength of a node is the sum of weights attached to its links. Nodes with more strength tend to have more power:

$$k_i^W = \sum_{j \in \Gamma(i)} w_{ij}$$

where $\Gamma(i)$ is the set of neighbors of a node i.

5. Betweenness Centrality

The betweenness centrality [19] of a node i is the proportion of all shortest paths between pairs of the other nodes that pass through this node. If we imagine information between nodes in the network and always taking the shortest possible path, then betweenness centrality measures the fraction of that information that will flow through a node. Betweenness centrality of node i is defined as:

$$b_i = \frac{1}{(n-1)(n-2)} \sum_{\substack{h,j \in N \\ h \neq j, h \neq i, j \neq i}} \frac{\rho_{hj}(i)}{\rho_{hj}}$$

where ρ_{hj} is the number of shortest paths between h and j, and $\rho_{hj}(i)$ is the number of shortest paths between h and j that pass through i.

6. Local Efficiency

The local efficiency (LE) is the global efficiency computed on node neighborhoods, and is related to the clustering coefficient. It is defined as:

$$E_{loc}^W = \frac{1}{n} \sum_{i \in N} \frac{\sum_{j,h \in N, j \neq i} \left(w_{ij} w_{ih} \left[d_{jh}^W (N_i) \right]^{-1} \right)^{1/3}}{k_i (k_i - 1)}$$

where $d_{jh}^W(N_i)$ is the length of the shortest path between nodes j and h that contains only neighbors of i [20].

E. Statistical Assessments

In order to test if the set of values of a graph index for each task follows the normal distribution, we used Kolmogorov–Smirnov test (K–S test), but the test rejected the null hypothesis at the 5% significance level.

So, Mann–Whitney U test was employed to identify the statistically significant differences between the graph indices of brain networks at different tasks. The difference was considered significant at $p < 0.05$.

III. RESULTS AND DISCUSSION

Taking into consideration the Fig.1, it seems that functional brain networks constructed from the two-digit multiplications tend to have high community structure and more communities (modules) than networks constructed from other tasks. Another important question on brain networks is how tightly clustered they are. We observe that clustering coefficient is higher during number looking and this indicates that there is higher clustering in these networks.

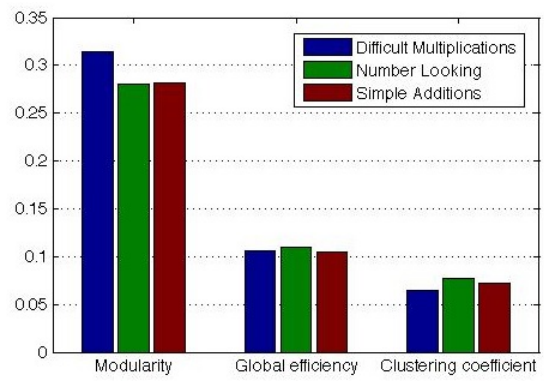

Fig. 1 **Global graph indices**. This figure depicts the values of global graph indices. Each bar represents the mean value of a graph index among the subjects at a specific task.

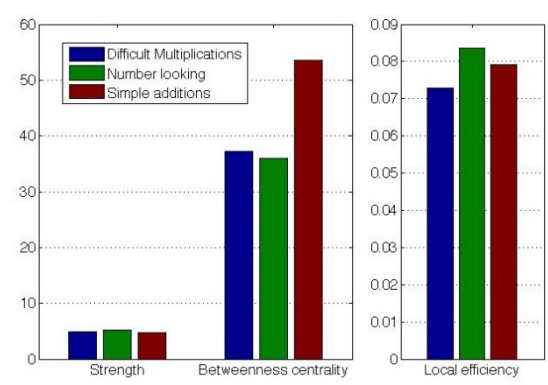

Fig. 2 **Local graph indices**. This figure depicts the values of local graph indices. We computed the average of each local index among the 57 nodes and then the average of this value among the subjects at each task.

Regarding the efficiency of the networks, subjects seem to have more efficient graphs during the task of number looking owing to the fact that both GE and LE are higher (Fig.1, Fig.2). Moreover, higher values of local efficiency suggest a larger level of internal organization and fault tolerance. So it seems that brain networks during number looking is more faults tolerant in contrast to those of other tasks.

As far as weights are concerned, it seems that on the average, nodes in network of NL tend to have more power than those of networks in other tasks. It is also clearly observable that betweenness centrality, a measure of a node's centrality in a network, is on the average higher in networks of SA.

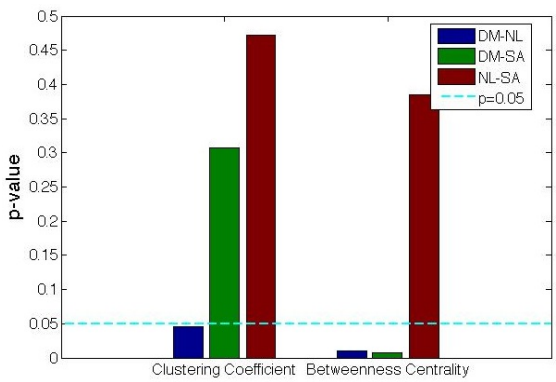

Fig. 3 In this figure we see a bar plot for p-values of Mann–Whitney U test. Each bar depicts the p-value from comparison of two sets, each containing 10 values of a specific graph index, one per subject. We observe that p-value from comparison of Difficult multiplications (DM) and Number Looking (NL) tasks is below the limits of 5%, for both Clustering Coefficient and Betweenness Centrality. In addition, p-value from comparison of (DM) and (SA) is also below 0.05 for Betweenness Centrality.

Finally, we found that there are statistically significant differences between DM and NL tasks for clustering coefficient and betweenness centrality. Furthermore, the networks of SA task had significant different betweenness centrality than networks of NL task.

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